

Using Python DataFrames for Advanced Database Applications

A DataFrame is a two-dimensional structure in which data is arrayed in labelled rows and columns. Pandas is a specialised Python library used effectively with DataFrames.



Data science tasks, nowadays, are not limited to the traditional data analysis with limited attributes and records. Currently, real-time data sets are huge in size with numerous attributes. Such data sets are very complex to evaluate using classical data analysis tools. For example, medical data sets integrate a number of attributes including information on symptoms, diagnosis, travel history, health parameters and many others that need to be evaluated. To deal with such types of data sets, assorted database query tools and programming languages are used.

Assorted tools and suites for database applications

The wide variety of tools and technologies available are used by data analysts and data scientists. This leads to effective prediction and visualisation prior to decision making.

Table 1 gives the commonly used tools for data exploration and analytics.

In addition to the above tools and suites, programming languages that can do high-performance statistical and database handling are also used. These include Python, Julia, R and many others. Depending on the application area and data set, one decides on whether to use a high-level

language or any other tool for data science. Data scientists use the tools or programming languages as per the outcomes they expect from the data science task.

Using Python Pandas

Python is a very powerful, multi-featured, cross-platform and high performance programming language that is used for multiple applications including data science, machine learning, deep learning, cyber security, cloud computing, grid computing, the Internet of Things, parallel computing, social media mining and many others.

To implement data science tasks, Python has different packages that enable predictive mining and analytics to be done with high accuracy. Python is enriched with multiple libraries and packages for data science based evaluations.

Working with DataFrames in Python Pandas for database applications

In Python, there is a specialised library called Pandas that is easy to use for data analysis and processing. The Pandas library executes very fast and integrates numerous functions for statistical analysis, data evaluation, database management, and many others.

Data scientists and analysts use the Pandas library as it has so many features that ensure high accuracy.

